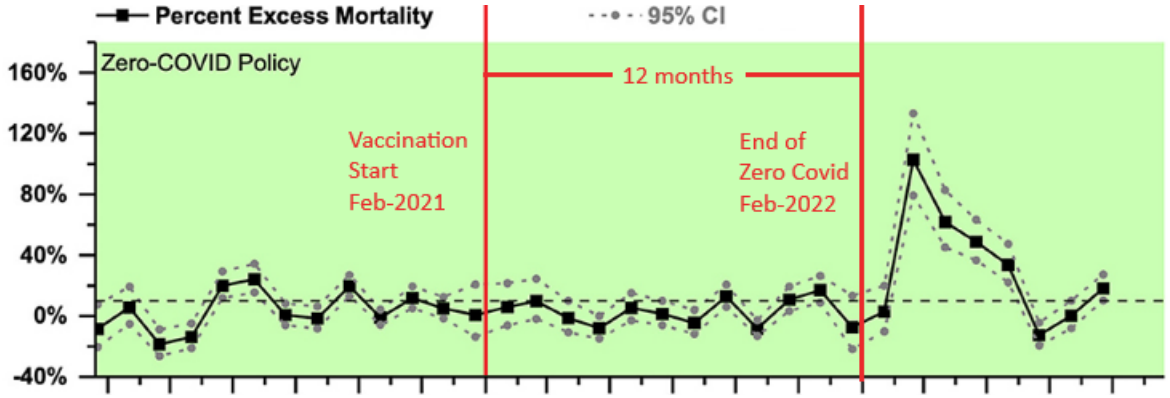
Argument 1: NCVIDs cannot exceed thousands of hypothesized deaths in the US

Zero Covid Countries

All six vaccinated countries that maintained near zero covid, show no ED until the end of the zero covid period (Hong Kong is shown below. Full details here). Note that this alone invalidates the Non-Covid Vaccine-Induced Deaths (NCVIDs) claim, which are 80% of the VIDs currently claimed by the opposition.



In the US we can see a similar result in highly vaccinated states during the months following initial vaccinations, when they experienced near zero covid. For example, in Massachusetts there were no EDs from March until August 2021 (until delta started).

No health authority detected a death signal

All major health authorities around the world have multiple processes to detect adverse effects of vaccines. These mechanisms have been proven to work multiple times, and more importantly, worked during the pandemic, detecting blood clots in Janssen and Astrazeneca (leading to their removal) and non-fatal myocarditis in mRNA (at a low rate that doesn't justify removal).

These were detected within months, and in parallel by multiple organizations around the world.
The opposition was unable to explain why thousands of institutions and scientists all over the world have suddenly decided to conspire to hide data only about mRNA vaccines and only for some types of adverse effects, hiding a death rate 1000x above other vaccines.

ED Attribution

The CDC provides detailed data about excess deaths, allowing breakdown by causes, ages, geographies, etc. We merged all ED causes during the pandemic into one table and found:

- Most EDs are covid deaths, both by medical designation and by observing how they perfectly track covid waves. There is negligible (or negative) ED during covid troughs.
- Additional EDs are attributed to overdoses, firearms and traffic accidents that increased during the pandemic, starting April 2020, long before vaccinations began.
- The remaining EDs are underreported covid deaths, as they track covid waves, but are not categorized as such.
  Note: The claim of many "deaths with covid" being wrongly categorized as covid deaths was found to be untrue.
- Once accounting for these causes, there are no unexplained EDs that could be NCVIDs.
- Interestingly, the number of non-covid EDs was actually lower in 2021 and 2022 relative to 2020 (before vaccines), further reducing the possibility of vaccine induced deaths hiding within the ED.

Death Cause	Excess Deaths		
	2020	**2021**	**2022**
Covid	348,653	470,402	263,401
Overdoses	26,899	41,799	43,041
Firearms	6,396	10,004	9,378
Traffic accidents	2,200	6,400	5,700
Non-Covid respiratory	-4330	-24761	-12255
Total	379,818	503,844	309,265
Reported EDs	453,810	530,876	327,658
Remaining ED	**73,992**	**27,032**	**18,393**

In summary, there are numerous ways VIDs could be detected with a sensitivity 100-1000x higher than the rate claimed by the opposition, and none were triggered. The only indicators of mass deaths the opposition brought forward are extrapolated anecdotes, misinterpretations of reporting systems like VAERS, and unsubstantiated attribution of covid EDs to vaccines.

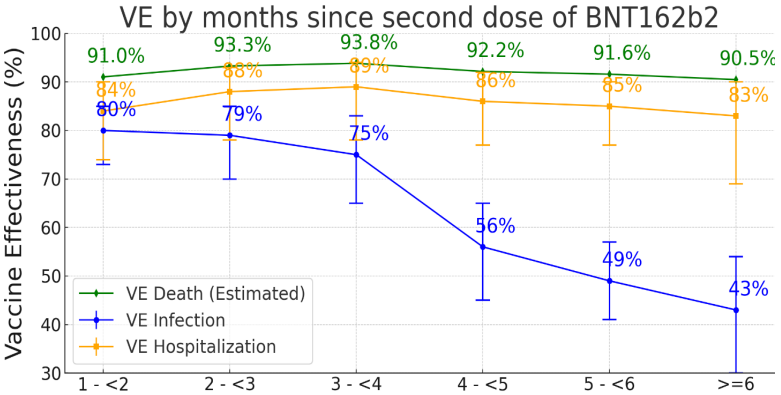
Argument 2: From every perspective examined, vaccination reduces covid deaths 10-fold, maintained for years.

Multiple studies, using different methodologies, in different populations and geographies, all consistently show the vaccinated died at a rate that is 90% lower than unvaccinated. Protection against infections did wane within months, but not against deaths - which is exactly the expected behavior of vaccinations and immune memory.

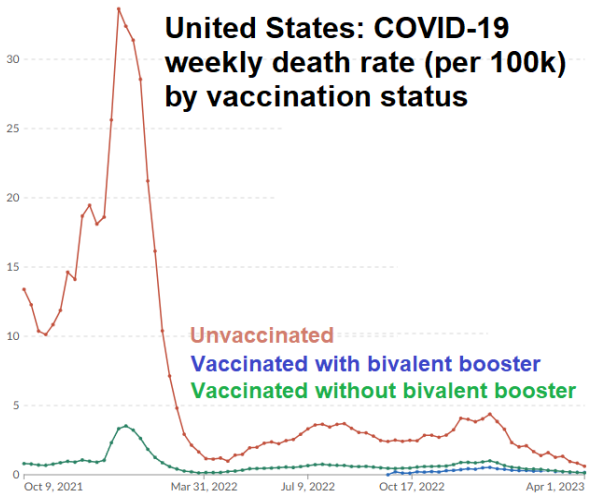
VE Studies

Some studies show lower VE_{death} over time, but only when failing to control for NI gained by the unvaccinated, making the relative protection (hybrid VE) drop. In any case, this lower hybrid VE was reached only after the omicron wave, after most of the benefit was already achieved.

We refer to multiple studies, including RCTs, test-control negative studies, cohort studies, and others, all showing the same result. A good example is the Kaiser Permanente study (chart to the right), which controlled for NI.



A Hong Kong VE study is especially interesting as it measured VE after a year of zero covid, and with low vaccination rates, thus reducing the effect of NI and HVE, and still found VE>90%. Hundreds of VE studies showing the same pattern are available here.

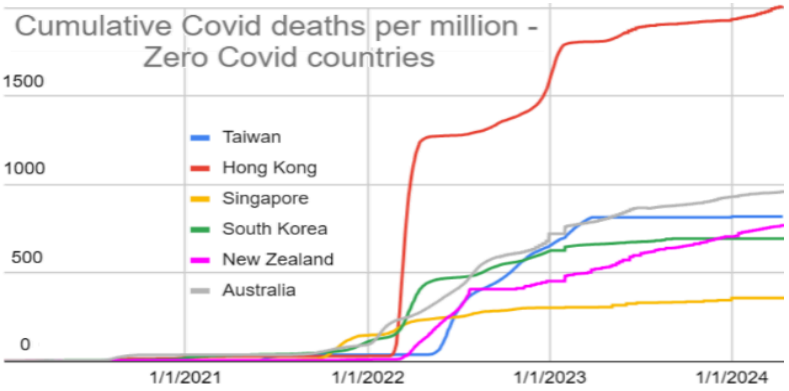


Covid Deaths by Vaccination Status

Across all geographies, charts of vaccinated vs unvaccinated covid deaths always show unvaccinated death rates far higher, as expected from a VE of 90%. The chart to the left shows this in the US, and the exact same pattern repeats in all mRNA-vaccinated countries we could find data for: Switzerland, Colombia, Finland (p. 14), Italy (p. 23), Malaysia, Northern Ireland (p.13), Netherlands (Fig. 1B), Israel.

While the opposition is claiming this is due to HVE, we demonstrate in the CZ data that HVE is at most a 3x effect that drops rapidly within weeks, so it hardly affects a long-term 10x reduction. Additionally, as shown below, the same effect is seen in methods not susceptible to HVE.

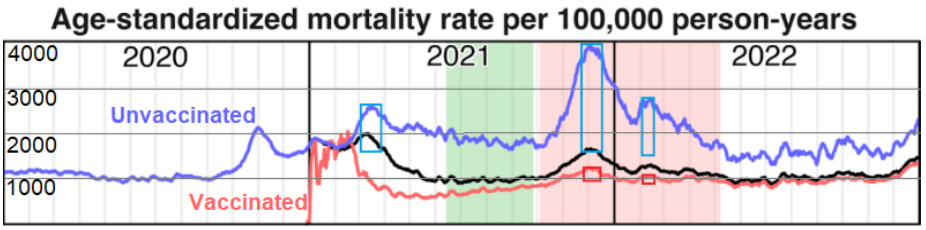
For example, comparing the covid deaths of the five zero-covid countries (on the right), shows Hong Kong, which had low vaccination rates for elderly, experiencing 2-5x more deaths, once zero covid ended. This is the expected effect at VE=90% and the opposite



of what is expected under Kirsch's hypothesis of no VE and vaccinations increasing covid deaths.

ACM by Vaccination Status

Measuring ACM rather than covid deaths amplifies HVE relative to the signal (as covid deaths are only 10-20% of total deaths). However, examining ACM over time and how it changes with covid waves allows controlling for most of the HVE.



In the current iteration we show the mortality rate by vaccination status in the Czech republic. While an HVE is clearly seen (with blue starting 3x higher than red, gradually dropping to 1.6x) the difference in how each group is affected by covid waves (blue and red boxes) indicates high efficacy. Specifically, the ratio of sizes of these boxes is 10x, exactly as expected from a VE of 90%.
Similarly, a comparison of republican and democrats shows little difference in ED prior to vaccination, and a significant worsening for the less-vaccinated Republicans after vaccination.
We plan to perform analyses of other ACM data in V5.

While many try to blame these consistent results on confounders, it is near impossible to find a set of confounders that account for all the evidence, namely:

- The RCTs show 95% efficacy.
- Real world results from early vaccinations (i.e. with a similar variant to the RCT and low NI) are nearly identical to the RCT.
- Different study designs, geographies, times, vaccination rates, variants etc. all show the same result.
- Population level studies (minimizing individuals' behavior confounders) show the same result. For example, when examining cases by age, we see older age groups, who were vaccinated earlier, had cases drop earlier than other age groups.

Of course, it is never possible to completely eliminate all confounders in such a complex system. However, the probability of having a set of confounders that explain these hundreds of data points, and does so in a way that cancels an effect of 10x is extremely low (and so far was not found).
Therefore, for all practical purposes, the statement that VE=90% should be considered true.

Argument 3: Modeling lives saved from VE=90% produces the same result as models that do not use VE at all: 500,000 to 1,000,000.

We provided a robust model for estimating lives saved that relies solely on the effect of VE_{death} on the vaccinated's first infection, thereby reducing a lot of the complexity of waning, variants etc. It estimates **627,240** lives saved.
We quoted many peer-reviewed models who use VE to reach similar numbers.
Note that such models are linearly affected by VE, so even if one wants to claim a 5x lower efficacy (i.e. VE=50%), it would still result in >300,000 lives saved.

We provide several models that **don't** use VE, and rather estimate lives saved solely from macro trends:

- The Czech ACM data above can be used to estimate lives saved by applying the unvaccinated's mortality rate to the vaccinated, while adjusting for HVE. This yields around **700,000** lives saved, when applied to the US.
- During delta, US states with higher vaccination rates experienced far fewer covid deaths. Extending the trend to zero vaccinations yields **400,000** lives saved in delta alone.
- After vaccinations, we see the death rate of older age groups drop more than in the young, due to their higher vaccination rate (the opposite of the predicted outcome under our opponent's hypothesis). By applying the age distribution of deaths before vaccinations to deaths post-vaccination, we calculated a lower bound for lives saved of **633,900**.

- Judge S provided a model extrapolating 2020 deaths into 2021 and 2022, if there were no vaccinations. This resulted in 2 million lives saved. Once replacing the model's ballpark estimates with data from studies, the result is **700,000**.

Each of these models relies on different assumptions, making their overall agreement very robust and reliable, as no single assumption can be attacked.

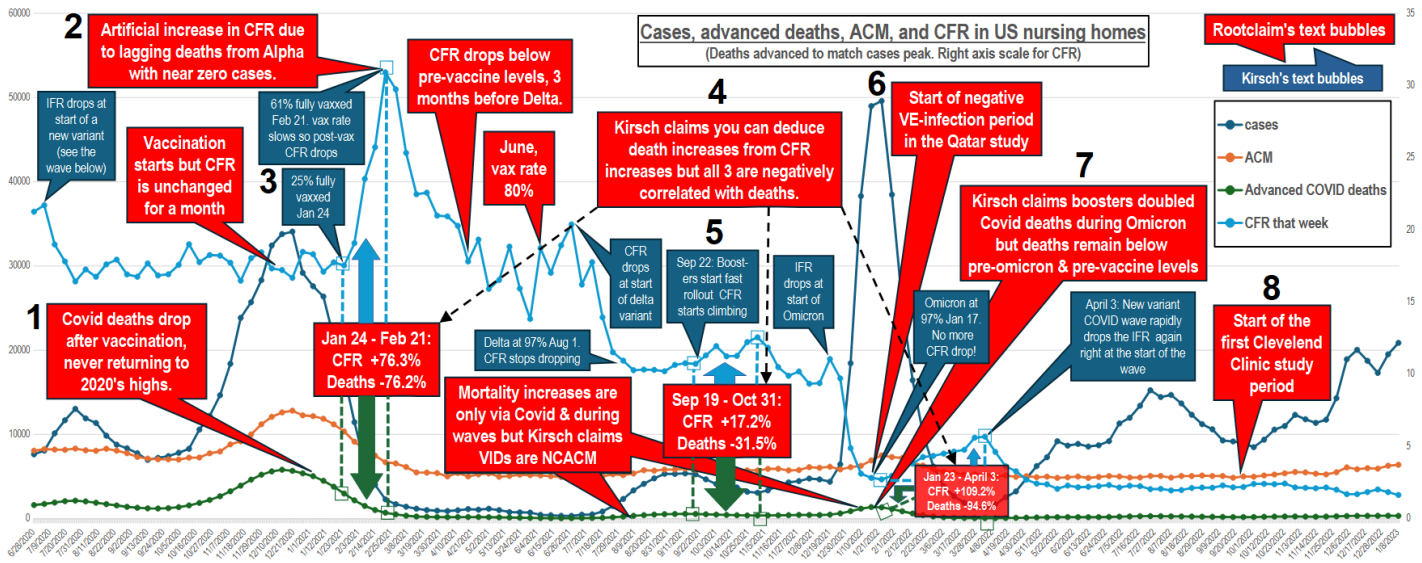
Argument 4: The opposition failed to produce a hypothesis

While not direct evidence, one of the strongest arguments is the failure of the opposition to produce any hypothesis that explains the evidence, or provide any evidence that counters our hypothesis. Thousands of people, some highly talented, spend enormous efforts sifting through mountains of data, trying to find any indication of deaths or inefficacy, but fail to produce anything substantial.

Convoluted Arguments

This forces them to produce highly convoluted arguments. For example, the opposition's main claim (assuming it doesn't change again) is that vaccines increased both CFR and cases. To reach that conclusion, they had to make a series of odd decisions:

- Estimating lives saved from CFR and cases separately - because measuring deaths directly clearly shows a benefit to vaccinated.
- Tracking CFR changes over time - because comparing vaccinated to unvaccinated clearly shows a benefit to vaccinated. This allows them to focus on statistical artifacts (lagging deaths when cases drop artificially raises CFR) and claim it's an increase in CFR caused by the vaccine (shown in bubble 2 below. The graph is Kirsch's US nursing home analysis, to which we added our comments in red).
- Using $VE_{infection}$ studies from long after most deaths have occurred - because at this time hybrid $VE_{infection}$ is low or temporarily negative, due to vaccinated "catching up" to the unvaccinated NI. If they were to use the more relevant earlier periods, it would show vaccines still provide protection against infection.
    - And even the studies that find negative $VE_{infection}$ clearly state it is likely due to NI. The Cleveland Clinic studies specifically show that when controlling for all confounders the pattern disappears.
- The biggest absurdity is that the splitting of deaths to cases and CFR is done at different times: The increase in CFR happens immediately after vaccinations (bubble 3), but instead of looking at the immediate impact on deaths (sharp drop - bubble 1), Kirsch deduces them indirectly by combining the CFR with the negative $VE_{infection}$ studies that start a year or more later: In Qatar (bubble 6) it is >7 months from second dose, which was given on May 2021, and Cleveland Clinic (bubble 8) starts 12-Sep-2022. No justification is given for these choices.
  We could use this methodology to cherry-pick the >90% drop in cases after vaccination, and the studies showing a 90% lower CFR in vaccinated, and merge them to claim VE=99%. There is no reason to prefer the opposition's calculation to this one (which is also wrong).



Internal Contradictions

Kirsch's lack of hypothesis is expected when using a faulty methodology that doesn't address outliers, biases and confounders. This produces contradictions that are missed because Kirsch doesn't check the internal consistency of his findings. Here are a few examples:

- What is the vaccines' impact on deaths?
  - Is it no impact, as Kirsch claims based on his Czech analysis?
    "*There was no vaccine benefit during COVID!! If we had injected the vaccinated cohort with saline, we would have expected to see the exact same result.*" (Czech record level data shows NO differential ACM benefit (V4)
  - Do they make you die 5x from Covid, as Kirsch claims based on his analysis of the RCTs:
    "*Pfizer "gold standard" RCT revealed that the CFR was over 5X higher for the vaccinated*" (V4).
- Who can detect VIDs and by what methods?
  - Can VIDs only be detected by a special autopsy, like Kirsch uses to explain why we should ignore no one finding the VIDs he claims?
    "Vaccine deaths are easy to miss unless the autopsy is very carefully done, which is pretty much never the case (the only exception being the 5 patients in Schwab 2022)." (V1)
  - Or can anyone reporting to VAERS do it, even if there's no autopsy or tests (which is almost always the case), like Kirsch claims as he assesses 100% of VAERS deaths are VIDs?
  - Or are the signs so obvious and common that "*73% of embalmers are consistently seeing these novel clots since the COVID vaccine rollout*" (V1)?
  - Or perhaps any survey respondent, with no autopsy, tests or even a medical background, knows when a death is a VID? (surveys such as Skidmore and Rasmussen, V1-4)
- What evidence should we rely on?
  - Can we only trust raw data, as Kirsch claims when he is confronted with hundreds of studies debunking his claims?
    "I don't trust studies since they all hide their data so I rely primarily on publicly available government data. We should be focusing our debate on this data and arguing what the data means. We don't need studies. We have the primary data. Plenty of it. Why ignore it?" (V4)
  - Yet, Kirsch quotes heavily the Skidmore study's estimate of 289,789 deaths, ignoring that it: a) uses a survey rather than raw government data, b) estimate deaths from a question about 'severe health problems', c) assumes a health problem **after** vaccination is caused by it.
    He similarly quotes several other studies, mostly non-peer-reviewed or retracted - none of which uses raw data.
- How many VIDs are there in the first 20 days?
  - Is it 28,000, as Kirsch claims based on a flawed autopsies paper (V4)?
    - "*14% of people who die within 20 days of vaccination were likely killed by the vaccine. Note that is a lower bound. [...] the total number of deaths [...] is probably around 20%*". So, VIDs in the first 20d = 20d ACM * vax rate * 20% = 27,750 VIDs.
  - Or is it 220,000, as is the result based on Kirsch's analysis of VAERS?
    - Kirsch *claims* VAERS' URF = 44. VAERS has 5k VIDs within 20d * 44 = 220,000.
  - This is an 8x error, meaning either VAERS URF is ~5 instead of 44, the 20% estimation based on the autopsies paper is wrong, VAERS reports are unreliable, or a combination of the three. Yet, all three are presented as separate and accurate calculations by Kirsch.

We emphasize these contradictions to point out that this is the expected result when many intelligent people try to force the evidence to support a conclusion that is 1000x different from the truth. A true hypothesis normally has the evidence aligned well, with occasional outliers, as seen with the consistent VE=90% found in every type of study, or the same number of lives saved calculated from very different types of data. But a wrong hypothesis is forced to cherry-pick just those outliers. And since mistakes and outliers occur for many different reasons, each produces very different results - resulting in inconsistencies, exactly as demonstrated in this section.